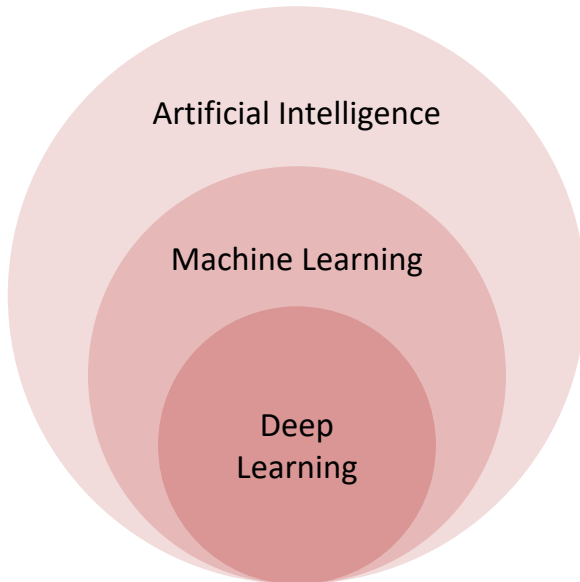


Introduction to Machine Learning

Introduction



Machine Learning (ML)

- A way to teach computers to learn from data
- Instead of giving exact rules, we give examples
- The computer finds patterns and makes decisions.
- For example, it can learn to:
 - Recognize faces or animals in pictures.
 - Recommend movies or songs you might like.
 - Predict prices of houses or *stock trends*.



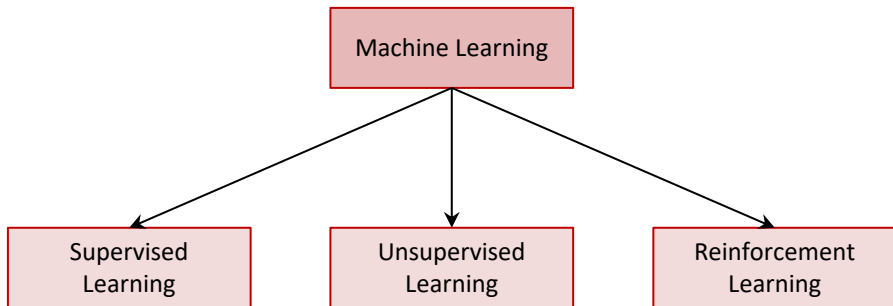
Steps of Machine Learning

Basic steps in Machine Learning:

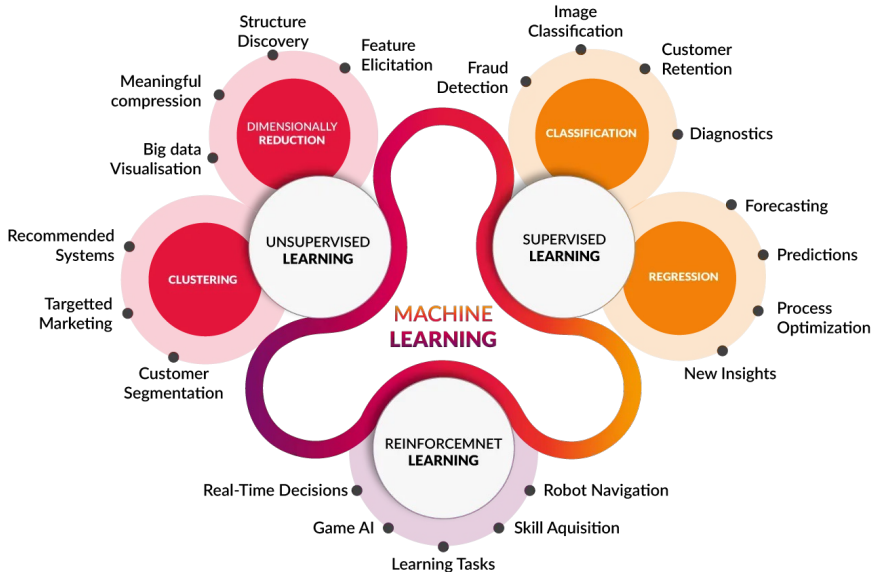
- Collect data
- Prepare data
- Choose a model
- Train the model
- Test the model
- Use it on new data

Question: How are the steps applied in the context of a stock price prediction model?

Types of learning



Types of learning



Supervised learning

Supervised learning is a type of machine learning where the computer learns from examples that include both the input (*data features*) and the correct answers (*data labels*).

It uses the training data to learn patterns and then makes predictions or decisions on new, unseen data.

Question: what are data features and labels in these applications?

- Email spam detection.
- Loan approval prediction.
- Medical diagnosis (like detecting diseases from scans).
- Customer churn prediction (who might stop using a service).

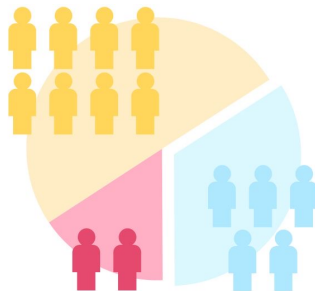
Unsupervised learning

Unsupervised learning is a type of machine learning where the computer learns from data without labels. It tries to find patterns, group things, or understand the structure of the data on its own.

Some examples of unsupervised learning:

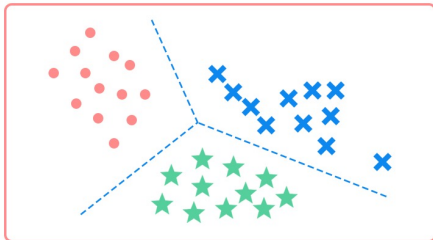
- Grouping customers by behavior.
- Finding unusual credit card activity.
- Organizing news articles by topic.

Question: what are data features of the above applications?



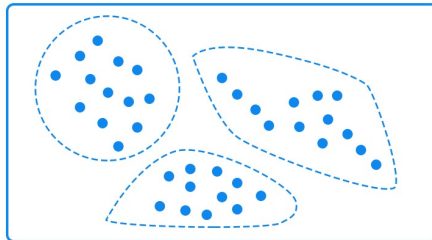
Supervised vs Unsupervised learning

Classification



Supervised learning

Clustering



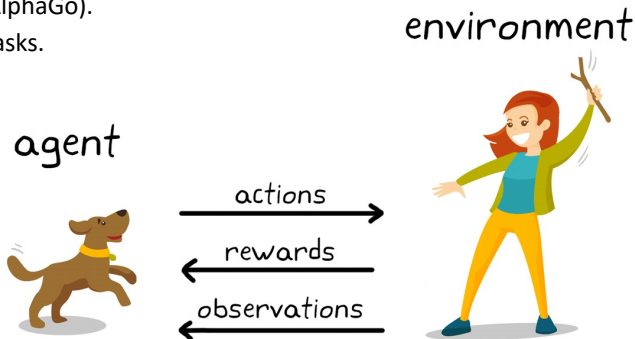
Unsupervised learning

Reinforcement learning

Reinforcement Learning (RL) is a type of machine learning where an agent learns to make decisions by interacting with an environment. The goal is to learn a policy that maximizes some notion of cumulative reward over time.

Some examples of RL:

- Game-playing AIs (like AlphaGo).
- Teaching robots to do tasks.



Reinforcement learning in dog training.

Supervised vs Unsupervised vs Reinforcement Learning

Type	Learn from	Used for
<i>Supervised Learning</i>	Labeled data	Prediction, classification
<i>Unsupervised Learning</i>	Unlabeled data	Grouping, pattern finding
<i>Reinforcement Learning</i>	Rewards and pennalties	Game, Robotics

Common Machine Learning tools

- Programming: Python
- Libraries: scikit-learn, TensorFlow
- Platforms: Google Colab, Kaggle Kernel



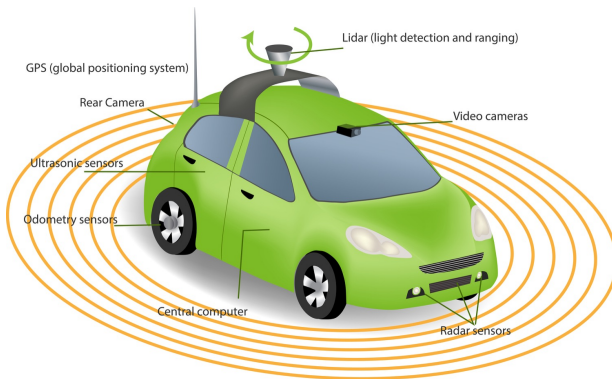
TensorFlow



kaggle

Real-life applications

- Self-driving cars
- Chatbots and virtual assistants
- Medical diagnosis
- Movie and product recommendations



Apply machine learning to the analysis of sensor data.

Summary

- ML is fun and useful.
- Opens doors to cool careers.
- You can build things that help people.



Q&A

Thank you